

RF and Microwave Engineering (EX752 / EX716)

Detailed PYQ

July 22, 2026

Notes

- Questions from Tribhuvan University, IOE (2069–2081 BS).
- Regular exam = **bold**, Back exam = regular font.
- BEX programme (EX752 / EG785) = normal font, BEI programme (EX716) = **this monospace style**.
- So: **72 Ash** = BEX Regular, 73 Ma = BEX Back, **79 Bh** = BEI Regular, 81 Ba = BEI Back.
- Months: Bh = Bhadra, Ma = Magh, Ash = Ashwin, Ba = Baishakh.
- Keywords: (I) = Importance/Need/Significance, (+) = Advantage, (-) = Disadvantage, +Fig = Figure, +Eg = Example, Vs = Compare/Differentiate.

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1 Introduction

(3 Hours)

1.1 Standard Frequency Bands & Microwave Applications

1. Classify microwave frequency bands + major applications [4] (69 Bh)
|→ How is the microwave frequency band classified by IEEE? [2+4] (80 Ba)
|→ Classify microwave frequency band + application in major areas [3+3] (**79 Bh**)
2. List areas of application of microwave systems [4] (**70 Bh, 70 Ma**)
|→ List uses of microwaves in communication systems + detailed protocols of any three [2+6] (81 Ba)
|→ Areas of application + detailed protocols of two microwave communication systems [3+3+4] (**80 Bh**)

1.2 Behaviour: Conventional Vs RF/Microwave Circuits

1. Lumped Vs distributed circuit analysis; uses of microwave bands [4+2] (73 Ma)
|→ Differentiate lumped and distributed circuit analysis (+ protocols of two M/W comm systems) [3+3+4] (**80 Bh**)
2. Microwave Vs conventional low-freq systems: working principles [4] (**70 Bh, 70 Ma**)
|→ Compare behaviour of circuits for low-freq/conventional Vs RF/microwave bands [3+3] (**79 Bh**)
|→ (+) (-) of microwaves over acoustic waves [5] (**72 Ash**)
|→ (+) (-): microwave Vs conventional low-freq [6] (71 Ma)
|→ (+) (-) of microwaves compared to lower frequencies [2+4] (80 Ba)
|→ Circuit at seismic band Vs RF/Microwave counterparts [8] (**75 Bh**)

2 RF and M/W Transmission Lines

(6 Hours)

2.1 Types of Transmission Lines

1. Microwave transmission lines Vs conventional low-freq lines [4] (69 Bh)
2. Microstrip Vs strip-lines [5] (71 Ma)

2.2 Smith Chart / Admittance (Immittance) Chart

1. Admittance chart [2] (72 Ash)
2. Immittance chart: sketch + list duality parameters [10] (75 Bh)
|→ Sketch of an Immittance Chart (short note) [4] (81 Ba)

2.3 Impedance Matching: Single Stub

1. Single stub shunt tuning (short & open): $Z_L = 80 + j100 \Omega$, microstrip [2+8] (72 Ash)
2. Single short/open shunt stub: $\Gamma_L = 0.5 \angle 51^\circ$, $Z_0 = 50 \Omega$ + Smith Chart [8] (74 Bh)

2.4 Impedance Matching: Double Stub

1. Double stub ($3\lambda/8$): $Z_L = 300 + j300 \Omega$, $Z_0 = 300 \Omega$ + Fig [10] (71 Bh)
2. Double stub ($3\lambda/8$): $Z_L = 80 + j180 \Omega$, $Z_0 = 100 \Omega$, 3 GHz + Fig [10] (70 Bh)
3. Double stub ($3\lambda/8$): $Z_L = 190 + j110 \Omega$, $Z_0 = 100 \Omega$, 10 GHz + Fig [10] (70 Ma)
4. Double stub: inductive load, $Z_0 = 50 \Omega$ + Fig [8] (73 Ma)
5. Double stub (short-circuit): $Z_L = 75 + j75 \Omega$, $Z_0 = 50 \Omega$, $3\lambda/8$ [3+15] (69 Bh)
6. Double stub (short-circuit): complex inductive load, $Z_0 = 78.0 \Omega$ + Fig [8+2] (73 Bh)
7. Double stub (short-circuit): $Z_L = 0.4 + j0.85$ (normalized), $Z_0 = 75 \Omega$ coaxial [10+2] (75 Bh)
8. Double stub shunt tuner (open stubs, $3\lambda/8$, first at 0.4λ): $Z_L = 60 - j80 \Omega$, 50Ω line; prepare S-matrix [8+2] (80 Ba)
9. Double stub (short-circuited coaxial): $Z_L = 109 + j120 \Omega$, 75Ω coaxial line; prepare S-matrix [8+2] (80 Bh)
10. Double stub: $\Gamma_L = 0.64 \angle 58^\circ$, $Z_0 = 100 \Omega$; prepare S-matrix of matched network [8+2] (81 Ba)
11. Double stub shunt tuner ($3\lambda/8$, first stub 0.01λ from load): 50Ω lossless line, load admittance $0.00813 + j0.0065 \mathcal{U}$ + Smith Chart; write S-parameter [8+2] (79 Bh)

3 RF and M/W Network Theory and Analysis

(4 Hours)

3.1 S-parameters: (I), Definition, Properties

1. S-parameters (I) in microwave network analysis [4] (**70 Bh**, 69 Bh)
|→ S-parameters (I); Vs h-parameters for network analysis [4+4] (70 Ma)
|→ S-parameters (I) [4] (**75 Bh**)
2. S-parameters for two-port network [4+5] (**70 Bh**)
|→ S-parameters for two-port network [4+6] (69 Bh)
|→ S-parameters for two-port network [4] (70 Ma)
|→ Why are S-parameters used in microwave network analysis? Derive S-parameters for a two-port network [2+6] (80 Ba)
|→ Need of S-parameter + properties; three-port network parameters using E-plane Tee example [4+4] (**80 Bh**)
3. S-parameters for three-port network [5+2] (**72 Ash**)
|→ S-matrix for 3-port network with +Eg [4+4] (73 Ma)
|→ Properties of 3-port network [4+4] (**73 Bh**)
4. Return loss & insertion loss [2] (**72 Ash**)
5. Why are S-parameters used at high frequencies? For $[S] = \begin{bmatrix} 0.4+j0.5 & j0.6 \\ j0.6 & 0.4-j0.5 \end{bmatrix}$: (a) reciprocal? (b) lossless? (c) return loss at input? (d) if input power is 5 W, reflected power? [3+1+1+1+2] (**79 Bh**)

3.2 Passive Microwave Devices: S-matrix Analysis

1. Identify + explain passive device from S-matrix (4-port) [8] (**74 Bh**)
2. Passive device from given S-matrix: judge condition + characteristics [6] (**75 Bh**)
3. Choose device to split power in half + explain properties [8] (73 Ma)
|→ Two radar transmitters, double power to antenna: solution + Fig, math [8] (**73 Bh**)
4. Prepare S-Matrices of a perfectly working E-Plane Tee, H-Plane Tee and a Duplexer, and explore their uses [8] (81 Ba)

4 RF/Microwave Components and Devices

(8 Hours)

4.1 Waveguides

1. Rectangular waveguide: TE mode field equations [7+3] (**70 Bh**)
 - |→ Rectangular waveguide: TM mode field equations + parameters [8+2] (71 Bh)
 - |→ Rectangular waveguide: TM mode field equations [7+3] (70 Ma) [10] (**74 Bh**)
 - |→ Rectangular waveguide: modes of propagation + critical parameters [8] (69 Bh)
 - |→ Field equations + basic characteristic equations of rectangular waveguide, TM_{11} mode [8] (81 Ba)
 - |→ TE_{mn} field equations of rectangular waveguide (short note) [4] (**80 Bh**)
2. TE_{10} Vs TE_{20} : cutoff frequency + dominant mode [2] (71 Bh)
 - |→ Show 1 GHz cannot propagate in TE_{10} , 5 cm wall separation [3] (**70 Bh**)
 - |→ Prove TM_{01} and TM_{10} modes do not exist [3] (70 Ma)
3. TE Vs TM mode: circular waveguide [10] (71 Ma)
 - |→ Fundamental field + characteristic equations of a circular waveguide, TE mode [10] (**79 Bh**)
4. Circular waveguide: TM wave field strength + dominant mode (TE & TM) [8+2] (**75 Bh**)
5. Dominant mode in waveguide [5] (**72 Ash**)
 - |→ What is dominant mode? Dominant modes for rectangular & circular waveguides. Does a rectangular waveguide of width 2.254 cm support 6 GHz? [1+1+2] (**80 Bh**)
6. For an air-filled rectangular waveguide (width 3 cm, $f = 6$ GHz, dominant mode): find cutoff frequency, cutoff wavelength, group velocity, phase velocity, guide wavelength and characteristic impedance. Also explain the four basic parameters of a directional coupler. [6+4] (80 Ba)

4.2 Coupling: Probes and Loops

1. PROBE-coupling Vs LOOP-coupling [5] (69 Bh)

4.3 Tee Junctions

1. Waveguide junctions; Magic Tee operational principles (S-parameters) [3+3] (**72 Ash**)
 - |→ Microwave magic tee [5] (71 Ma, 71 Bh)
 - |→ Draw a neat diagram of a magic tee and derive its S-parameter [4] (**80 Bh**)
 - |→ Microwave Magic Tee (short note) [4] (**79 Bh**)
2. E-plane tee Vs H-plane tee [5] (69 Bh)
 - |→ E-plane Tee (short note) [4] (80 Ba)
3. Hybrid tee [8] (70 Ma)
4. 3-port directional coupler (S-parameters analysis) [4] (**75 Bh**)

4.4 Directional Couplers

1. Directional couplers [6] (73 Bh)

4.5 Other Devices (Gunn diode, MASER)

1. Gunn-diode (short note) [4] (**79 Bh**)
2. MASER (short note) [4] (80 Ba)

4.6 Resonators and Circulators

1. Circulators
2. Microwave cavity resonators

[5] (**72 Ash**, 70 Bh, 73 Ma)
[6] (**74 Bh**)

4.7 Microstrips

1. Microstrips

[5] (69 Bh)

5 Microwave Generators

(5 Hours)

5.1 Transit-time Effect

1. Transit-time effect + two-cavity klystron construction & operation [2+8] (70 Ma)

5.2 Klystrons

1. Bunching effect (density modulation) [2] (**72 Ash**, 71 Bh, 70 Bh, 70 Ma)
2. Multi-cavity klystron oscillator working principle [2+7] (**72 Ash**) [2+8] (71 Bh)
|→ Two-cavity klystron [5] (69 Bh)
3. Velocity modulation + bunching in multicavity reflex klystron + Fig [10] (**75 Bh**)

5.3 TWT and Magnetrons

1. Cavity magnetron: construction + operational features [2+8] (**70 Bh**) [10] (71 Ma)
|→ Sketch a cross-sectional magnetron with 45 phase shift among adjacent cavities; explain its functioning as a power amplifier + bunching effect [6+2] (80 Ba)
|→ Working principles of a cross-field magnetron with 60 phase differences among adjacent cavities [8] (81 Ba)
2. What is bunching effect? Describe the construction and operation of Traveling Wave Tube (TWT) [2+6] (**80 Bh**)
|→ How is the output of conventional tubes reduced at microwaves (inter-electrode capacitance, lead inductance, transit time)? Construction & working of TWT [2+6] (**79 Bh**)

5.4 Backward Wave Oscillator (BWO) and LNA

1. BWO: slow backward wave structure, construction + principle [2+8] (73 Ma) [5] (**74 Bh**)
2. LNA: construction + working principle [2+6] (73 Bh)

6 RF Design Practices

(10 Hours)

6.1 RF Filter: Insertion Loss Method & Microstrip Implementation

1. LNA cavity device – insertion loss method for filter design [5] (72 Ash)
2. Insertion loss method for filter design [5] (75 Bh) [8] (73 Bh)
|→ Butterworth & Chebyshev: common to two-port prototype filter (insertion loss) [4] (70 Bh, 70 Ma)
|→ Describe insertion loss method, illustrate passive LPF using μ -strip [8+2] (73 Ma)
3. Microwave filter design procedures [5] (71 Bh)
|→ Designing steps + passive HPF + Eg using microstrips [6+4] (74 Bh)
|→ Steps of microwave filter designing; sketch a double-pad T-type passive filter using microstrip [7+3] (81 Ba)
4. LPF using microstrip + how prototyped [3+5] (72 Ash)
|→ Why is the insertion loss technique used to design microwave filters? Sketch a double-section shunt-arm microwave LPF using microstrips [3+5] (80 Ba)
|→ How is a LPF prototyped based on Butterworth/Chebyshev and converted to other filters? Implement a 2nd-order HPF π -section using microstrips [10] (80 Bh)
|→ How is a LPF prototype (Butterworth) designed using the insertion loss method? Implement a LPF π -section using microstrips [6+2] (79 Bh)

6.2 RF Amplifier: Theory, Stability, Design

1. Amplifier Vs oscillator circuit: stability factor [5] (71 Ma)
2. Stability + max gain (conditional stable, unilateral): $S_{11}=0.894\angle-60.6$, $S_{12}=0.020\angle62.4$, $S_{21}=3.122\angle123.6$, $S_{22}=0.781\angle-27.6$ [5+5] (72 Ash, 70 Ma)
|→ Same S-parameters: determine stability + compare max power gains for bilateral & unilateral modes [5+5] (80 Ba, 79 Bh)
3. Stability + max gain (bilateral + unilateral): $S_{11}=0.55\angle150$, $S_{12}=0.04\angle20$, $S_{21}=2.82\angle180$, $S_{22}=0.45\angle-30$ [10] (71 Bh)
|→ Max gains of both bilateral & unilateral models; sketch flowchart of microwave amplifier designing procedure [6+3+3] (81 Ba)
4. Stability + max gain (bilateral + unilateral + Γ_{in} , Γ_{out} on Smith Chart): $S_{11}=0.656\angle146.7$, $S_{12}=0.122\angle46.1$, $S_{21}=2.30\angle44.7$, $S_{22}=0.172\angle-117.1$ [4+4+4+4] (74 Bh)
|→ Max gain (bilateral + unilateral): same S-parameters [10] (70 Bh, 71 Ma)
5. GaAs FET at 4 GHz: $S_{11}=0.72\angle-116$, $S_{12}=0.03\angle57$, $S_{21}=2.60\angle76$, $S_{22}=0.73\angle-54$ [18] (69 Bh)
6. GaAs MESFET at 10 GHz: $S_{11}=0.55\angle-160$, $S_{12}=0.04\angle10$, $S_{21}=4.82\angle180$, $S_{22}=0.45\angle-30$; determine stability + compare max power gains for unilateral & bilateral [10] (80 Bh)
7. Stability + max gain (bilateral + unilateral): $S_{11}=0.64\angle-169$, $S_{12}=0.03\angle50$, $S_{21}=10.11\angle91$, $S_{22}=0.22\angle-82$ [12] (73 Ma)
8. Flowchart for microwave amplifier design (GaAsFET) [4] (74 Bh)
|→ Flowchart + stability of $C_S=1.15\angle10$, $R_S=0.85$, $C_L=1.10\angle80$, $R_L=1.10$ [5+5] (71 Bh)
9. Stability synthesis from Smith Chart (conditional stability) [8] (74 Bh) [10] (73 Bh)
|→ Stability analysis of a microwave amplifier (short note) [4] (81 Ba)

6.3 Oscillator and Mixer Theory

1. Mixer theory [5] (70 Bh)

7 Microwave Antennas and Propagation

(3 Hours)

7.1 RF/Microwave Radiation, Safety Practices and Standards

1. Microwave radiation hazards (radiation fields) [8] (69 Bh) [5] (71 Ma, **72 Ash**)
 - |→ RF radiation hazards + safety standards [8] (70 Ma)
 - |→ RF/MW radiation hazards + safety practices [6] (73 Bh, **74 Bh**) [3+3] (**79 Bh**)
 - |→ Effect of SAR as microwave radiation hazard [6] (**74 Bh**)
 - |→ How microwave radiation becomes hazardous to human body [5] (**75 Bh**)
 - |→ Discuss RF radiation fields to public exposers; international standards and recommended (SARPs) practices to be safe [3+3] (80 Ba)
 - |→ Classify microwave propagation fields of a GSM BTS + radiation parameters; what non-ionizing SAR is and how it becomes hazardous [4+4] (**81 Ba**)
 - |→ HERP (short note) [4] (**80 Bh**)
2. Microwave radiation fields [5] (71 Ma)
3. Antenna as DUT: choose measurement tool + working principle [8] (**74 Bh**)

8 RF/Microwave Measurements

(6 Hours)

8.1 Power Measurement: VSWR and Bolometry

1. Standing waves + microwave power: VSWR meter + low power measurement [2+8] (**72 Ash**)
|→ VSWR meter + bolometry [4+6] (**70 Bh**)
|→ List major RF/MW measurement parameters; how is the VSWR of a microwave transmitter (VSWR > 10) measured? [2+6] (**79 Bh**)
2. Microwave measurements Vs low-frequency measurements [3] (71 Bh)
3. Major microwave measurement parameters + low power measurement device [8] (73 Bh)
4. Power measurement using calorimeter wattmeter; limitations of a single-bridge bolometer [6+2] (80 Ba)
|→ Power measurement + working principle of the double-channel Bolometer Bridge method [8] (**80 Bh**)
|→ Double-channel Bolometer (short note) [4] (81 Ba)
5. Spectrum Analyzer (short note) [4] (**80 Bh**)

8.2 Calorimeter Methods

1. Static calorimeter working principle [5] (71 Ma) [7] (71 Bh)
2. Calorimetry in microwave; circulating Vs flow calorimetry [2+8] (70 Ma)
|→ Circulating calorimeter (power measurement) [5] (**75 Bh**)

8.3 Network Analyzer and Two-Port Models

1. Two-port network model + derive parameters [10] (71 Bh)
2. Network analyzer working principle [8] (73 Ma)